

Data Science

CSE558

Recap

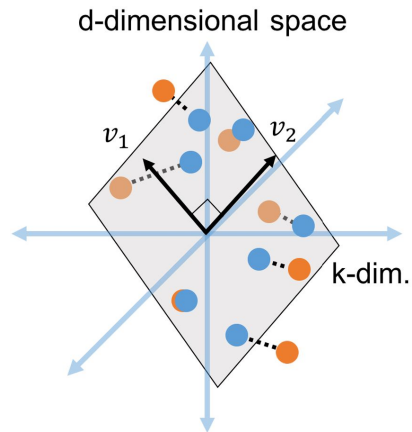
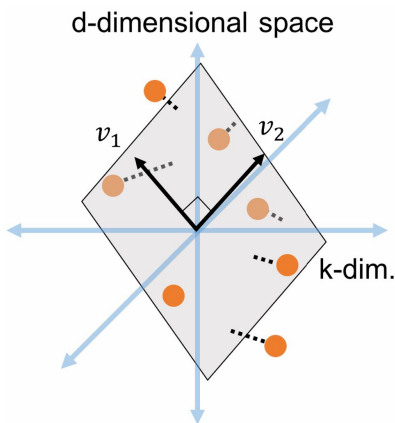
- Fast Linear Regression
- Low Rank Approximation

Overview

- Low Rank Approximation
- Fast Matrix Multiplication

Low Rank Approximation

Suppose $x_1, x_2, \dots, x_n \in \mathbb{R}^d$ close to a k -dimensional space.



For every x_i a distorted embedding $c_i \in \mathbb{R}^k$ is a good approximation.

- Compute embedding
- For every x_i compute approximate $y_i \in \mathbb{R}^d$ (Low Rank Approximation)

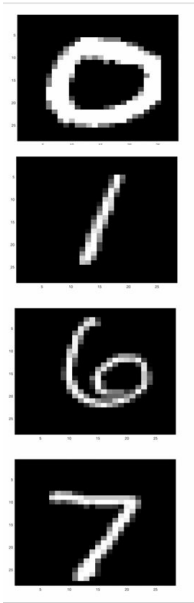
Low Rank Approximation

$$X = I \cdot X = (A + B) \cdot X$$

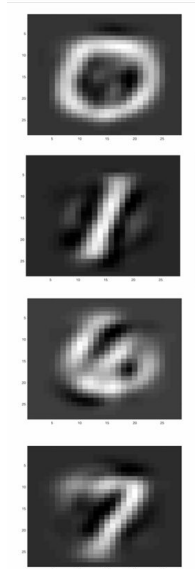
- Idempotent: $B \cdot B = B$

Applications

28 x 28 pixels



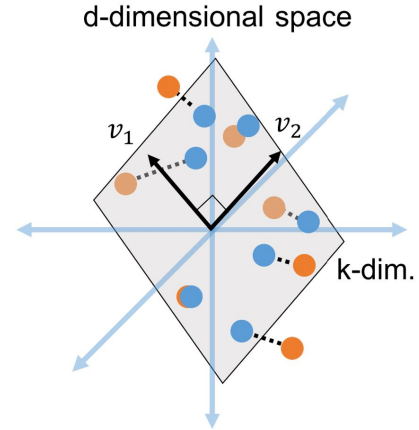
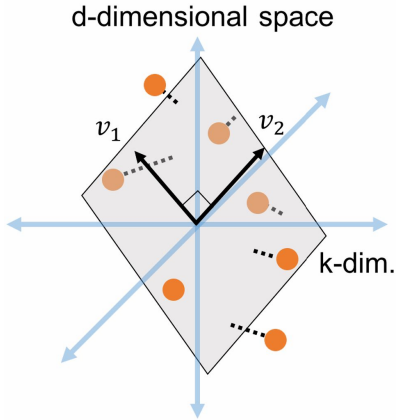
16 pixels



Linearly dependent variable

bedrooms	bathrooms	sq.ft.	floors	list price	sale price
2	2	1800	2	200,000	195,000
4	2.5	2700	1	300,000	310,000
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
5	3.5	3600	3	450,000	450,000

Question



How to find the k -dimensional subspace and then its orthogonal basis?

Best Fit Subspace

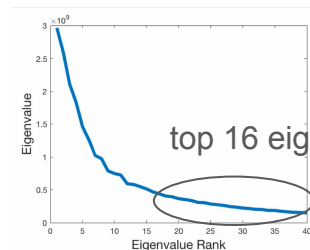
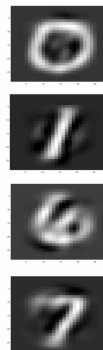
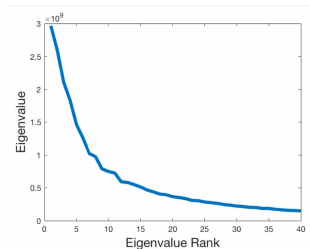
Suppose $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \in \mathbb{R}^d$ close to a k -dimensional space.

$$\arg \min_{\text{orthonormal } \mathbf{V} \in \mathbb{R}^{d \times k}} \|\mathbf{X} - \mathbf{XV}\mathbf{V}^T\|_F^2 = \sum_{i,j} (\mathbf{x}_{i,j} - (\mathbf{XV}\mathbf{V}^T)_{i,j})^2$$

$$\arg \max_{\text{orthonormal } \mathbf{V} \in \mathbb{R}^{d \times k}} \|\mathbf{XV}\|_F^2$$

Greedy algorithm computes the orthonormal $\mathbf{V}$.

Spectrum Analysis

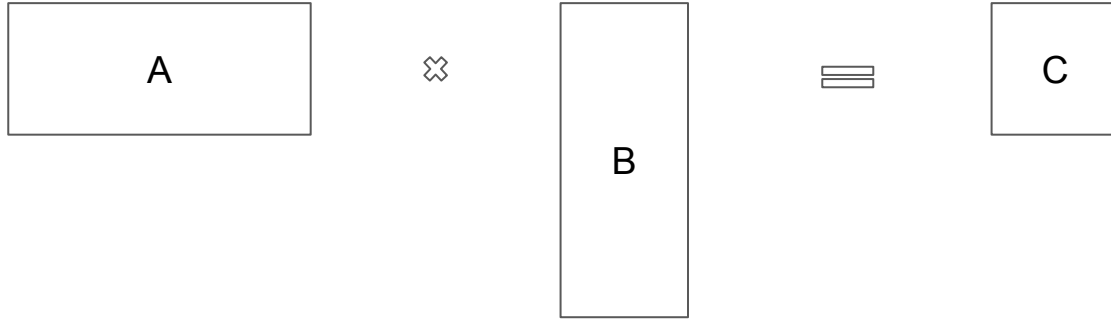


$$\|X - XV_k V_k^T\|_F^2 = \sum_{i=k+1}^d \lambda_i(X^T X)$$

Overview

- Low Rank Approximation
- Fast Matrix Multiplication

Matrix Multiplication



Expensive!

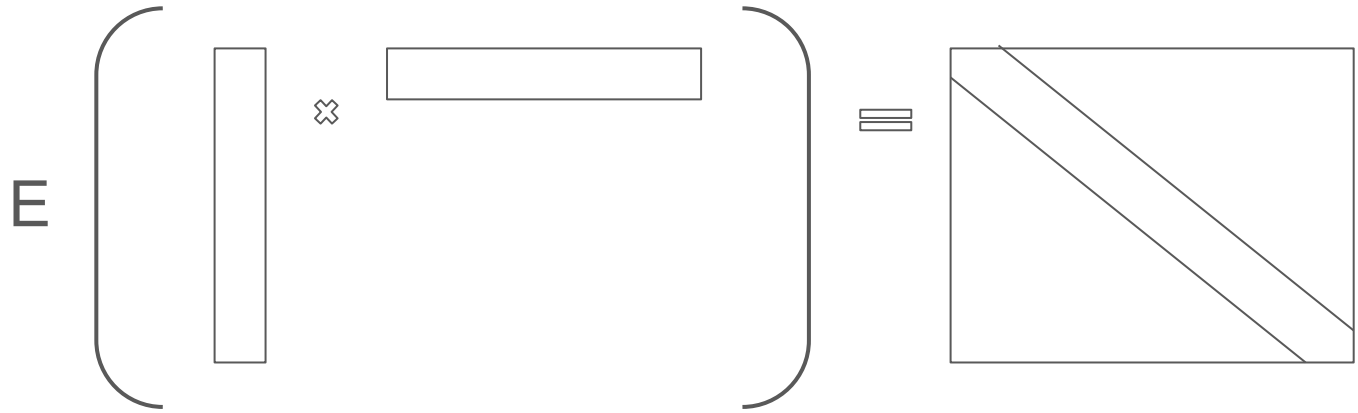
Matrix Multiplication



Fast matrix multiplication by projection.

Random vector from $N(0, I)$.

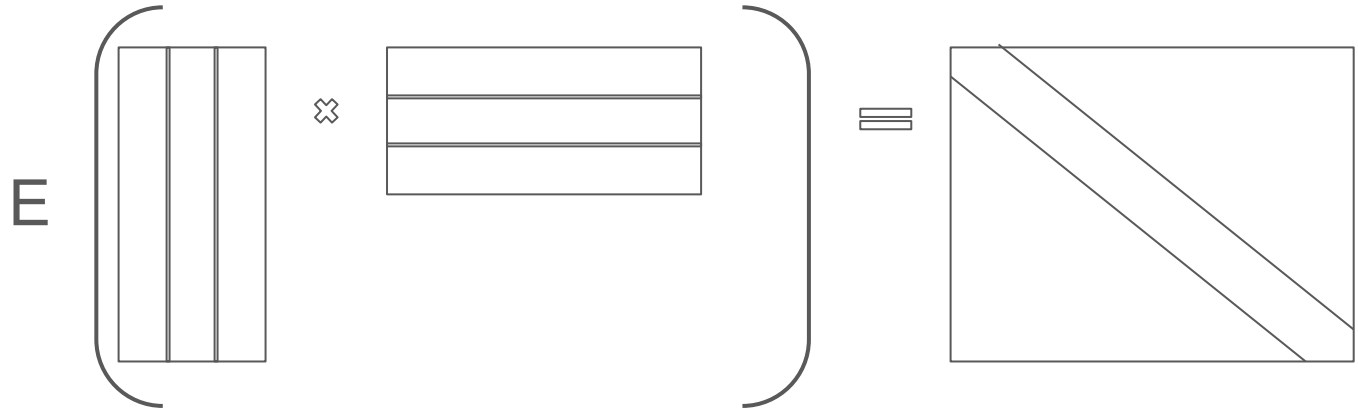
Matrix Multiplication



Fast matrix multiplication by projection.

Random vector from $N(0, I)$.

Matrix Multiplication

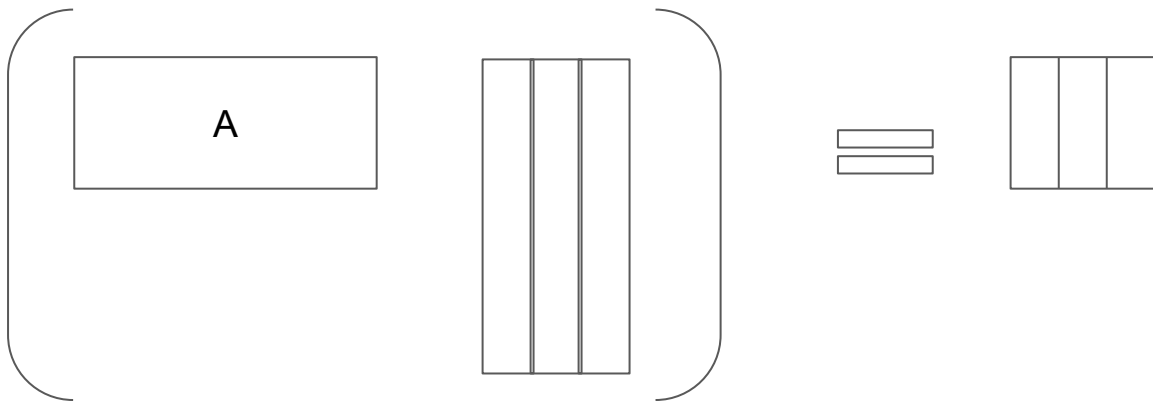


Fast matrix multiplication by projection.

Random vector from $N(0, I/3)$.

Problem with Random Projection

- Structural Loss



Sparse

Interpretable features.

Dense

features not interpretable.